

RICHA SARAF

www.linkedin.com/in/richasaraf22/ +1 (571)-765-0005 | richasaraf95@gmail.com

SUMMARY

A data-driven and results-oriented professional with a passion for building efficient data pipelines. Demonstrated proficiency in cloud platforms (GCP, AWS, Azure) and utilize Python, SQL, and R to tackle complex data challenges. Spearheaded capstone project bridging the gap between research and real-world application by implementing GenAI for medical image analysis and disease progression forecasting.

WORK EXPERIENCE

Jury Systems Incorporated

Simi Valley, CA

Data Engineer Intern

June 2023 – August 2023

- Applied robust data quality management and continuous monitoring, reducing data discrepancies by 35% in jury systems across 50+ counties in California, Florida, and Arizona.
- Executed ETL processes to seamlessly migrate and integrate data from legacy systems into advanced jury management solutions.
- Designed efficient ETL workflows using Airflow's advanced scheduling and orchestration, reducing data migration time by 70% across 10 counties and ensuring continuity of court operations during the transition period.
- Optimized jury selection and scheduling, achieving a 25% efficiency improvement across all serviced counties.

TheMathCompany

Bangalore, IN

Data Engineer | Social media analytics of a leading beverage company

February 2022-July 2022

- Integrated diverse media sources and automated raw data storage to Google Cloud Platform (GCP), cleaning data streams from 18 APIs via ELT pipeline. Resulted in a platform for marketing teams to compare source metrics, reducing analysis time by 95% and manual work by 67%.
- Developed OAuth2 Flow to access data from platforms including Facebook, YouTube Ads, Google Search Ads, Snapchat, and TikTok, and developed a token refresh flow to automate token renewal, enhancing operational efficiency.
- Automated the entire process using Airflow and orchestrated data transformation and ingestion into Snowflake using Google Cloud Composer. Deployed a monitoring system that reduced error resolution by 50% through pipeline failure notifications.

Data Engineer | Footfall prediction in airline lounges for a leading airlines

September 2021-February 2022

- Conducted comprehensive market research and data procurement, addressing data gaps through preprocessing techniques to develop an optimal solution for forecasting passenger footfall for Leading Airlines.
- Utilized extensive Exploratory Data Analysis (EDA) and hypothesis testing on a dataset of approximately 42,000 rows, leveraging Tableau to facilitate robust analysis.
- Leveraged ACF, PACF and feature selection before developing a sales blueprint integrating EDA findings and executed 3 advanced machine learning techniques (AUTO ARIMA, FB PROPHET, UCM) achieving 92% accuracy.
- Incorporated models into an in-house application (Co.dx), designing backend architecture with Azure Blob Storage and GitLab and ensuring seamless integration and management using Co.dx.

ACADEMIC PROJECTS

Medical Image Sequencing using Generative AI

Jan 2024 – May 2024

- Processed approximately 48,000 input images (19GB of data) as RGB, ensuring compatibility and leveraging color information for comprehensive analysis of lung adenocarcinoma.
- Instituted Vision Transformer and Variational Autoencoder models, elevating predictive capabilities to boost medical imaging analysis for effective treatment planning, with the model predicting outcomes up to 2 weeks in the future.
- Leveraged GitHub for version control and Utilized Hopper cluster for compute power.
- Conducted hyperparameter tuning using Pydicom, optimizing feature dimension to 512, learning rate to 1e-3, and image size from 512 x 512 to 224 x 224, achieving the highest Structural Similarity Index (SSIM) of 0.64.

Liquor Sales Prediction: Advanced Sales Forecasting and Supply Chain Optimization

June 2024 - Present

- Developing a robust framework to identify, rectify, and continuously monitor data quality issues, ensuring the reliability of historical sales data (~3M rows and 24 columns) for accurate forecasting
- Optimizing supply chain by aligning ongoing sales forecasts with customer demand, incorporating seasonal and regional variations to optimize inventory levels.
- Implementing and maintaining a scalable MLOps pipeline for continuous automation of data ingestion, preprocessing, model training, and deployment, ensuring ongoing model updates and seamless integration with existing systems.

CERTIFICATIONS

- Accredited in Time Series Analysis in **R** from DataCamp. November 2023
- Certified in Cluster Analysis with R from Data Camp. October 2023
- Certified for **AWS** Intro to Data Pipeline October 2023
- DELF B1 Certified in French language by Alliance Française in 2016. November 2016

EDUCATION

George Mason University (Present)

Fairfax, VA

Master's in Data Analytics Engineering

May 2024

RN Shetty Institute of Technology (2017-2021)

Bangalore, IN

Bachelor of Engineering in Information Science

June 2021

TECHNICAL SKILLS

- Programming Languages/Version Control/CRM Tools: Python (NumPy, Pandas, Matplotlib, torchvision, cv2, os, transformers, gc, skimage.metrics, timm), SQL, R, Git, YouTrack, Jira, Salesforce
- Databases/Data Visualization and analysis: MySQL, Oracle, Snowflake, Tableau, Power BI, Microsoft Excel, Apache Zeppelin
- Cloud Platform/Data Engineering and Processing: Google Cloud Platform, Apache Airflow, Apache Spark, Docker, Hopper Cluster